

Alexa's Conjecture on Primality

OPG-37413

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Definition Let r_i be the unique integer (with respect to a fixed $p \in \mathbb{N}$) such that

$$(2i + 1)^{p-1} \equiv r_i \pmod{p} \text{ and } 0 \leq r_i < p.$$

Conjecture A natural number $p \geq 8$ is a prime iff

$$\sum_{i=1}^{\left\lfloor \frac{\sqrt[3]{p}}{2} \right\rfloor} r_i = \left\lfloor \frac{\sqrt[3]{p}}{2} \right\rfloor$$

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>